Eliminating Child Care Deserts in New York State through Optimization

Group 29

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Abstract

This project addresses the critical issue of child care deserts across New York State using optimization modeling. By integrating demographic, economic, and spatial data, the study aims to identify optimal strategies for expanding and building child care facilities within a limited budget. Three sequential tasks were completed: (1) estimating the minimum funding required to eliminate deserts, (2) introducing realistic expansion and location constraints, and (3) incorporating fairness and social coverage considerations. Mixed-integer linear programming (MILP) models were implemented in Python with Gurobi to determine cost-effective and equitable policy solutions for the New York State government.

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1. Introduction

The issue of childcare deserts in New York State is analyzed and tackled in this report using optimization modeling. Childcare, or day care, refers to the licensed supervision and care of infants (2 weeks to 12 years old). A *childcare desert* refers to regions where the existing licensed facilities are insufficient to satisfy the demand for care. This study focuses on New York State and proposes both expansion and construction strategies to meet the childcare demand under a state budget of $100 million.

The model is based on publicly available datasets describing existing childcare facilities and demographic indicators. Data at the ZIP code level include population of children by age, average individual income, employment rate, and potential available locations for new facilities. The problem is formulated as a **Mixed-Integer Linear Program (MILP)** and implemented in Python using the Gurobi solver to obtain an optimal solution.

The study proceeds through three stages: (1) estimating the minimum total funding required to eliminate childcare deserts (Task 1), (2) introducing more realistic cost functions and spatial constraints (Task 2), and (3) incorporating fairness considerations to ensure equity across regions (Task 3). The overall goal is to propose optimal and effective strategies for facility placement and expansion to address the childcare crisis in New York State.

2. Background Information

In this section, we provide contextual and statistical background about childcare deserts in New York State and highlight the importance of addressing this issue through optimization modeling.

Childcare (or day care) refers to the licensed care and supervision of children between the ages of two weeks and twelve years old [1]. This service is essential for working families, as it provides a safe environment for children's development and socialization while parents are employed [2]. In New York State, a significant number of regions have been identified as *childcare deserts*, where the demand for licensed childcare slots exceeds the available supply. According to national estimates, more than half of American children live in such areas characterized by limited childcare capacity, high employment rates, and low average income [3].

High- and Normal-Demand Classification

In this study, **high-demand** ZIP codes are defined as those with an employment rate above 60% or an average annual income below $60,000. A ZIP code is classified as a childcare desert if the number of licensed slots for children aged 0–12 fails to meet the minimum threshold:

$$\text{High-demand: } \frac{\text{slots}}{\text{population}_{0-12}} \leq 0.5, \quad \text{Normal-demand: } \frac{\text{slots}}{\text{population}_{0-12}} \leq \tfrac{1}{3}.$$

Furthermore, due to the greater care needs of younger children, each region must provide enough capacity to serve at least two-thirds of its 0–5 population.

Facility Types and Cost Structure

Three discrete sizes for new childcare facilities are considered, as shown in Table 1. Existing facilities may be expanded by up to 120% of their current capacity, or by no more than 500 additional slots. Expanding a facility by 100% or more triggers a fixed baseline cost of $20,000 plus $200 per existing slot, while each new slot for children aged 0–5 requires an additional equipment cost of $100.

Facility Type	Total Slots	0–5 Slots	Build Cost ($)
100 slots (Small)	100	50	65,000
200 slots (Medium)	200	100	95,000
400 slots (Large)	400	200	115,000

Table 1: Facility sizes, 0–5 capacities, and construction costs.

To better capture the true expansion cost, a tiered cost function is introduced in later tasks. Expansions beyond 20% are disallowed, and larger expansions incur higher marginal costs according to:

$$c_{\text{exp}} = \begin{cases} (20{,}000 + 200n_f)\,\frac{x_f}{n_f}, & 0 < \frac{x_f}{n_f} \leq 0.10, \\ (20{,}000 + 400n_f)\,\frac{x_f}{n_f}, & 0.10 < \frac{x_f}{n_f} \leq 0.15, \\ (20{,}000 + 1000n_f)\,\frac{x_f}{n_f}, & 0.15 < \frac{x_f}{n_f} \leq 0.20, \end{cases}$$

where n_f is the existing capacity and x_f the number of new slots added at facility f.

Socioeconomic Context

Childcare deserts disproportionately affect low-income and rural communities. In New York State, approximately 60% of the population resides in childcare deserts [5]. These areas are predominantly rural, with three out of five rural communities lacking sufficient licensed facilities to meet demand. Low-income and Hispanic/Latino families are particularly affected, often forcing parents—especially mothers—to leave the workforce to care for children, leading to economic vulnerability and reduced quality of life [6].

Although the state has taken measures, such as the Universal Pre-K initiative and increased funding since 2022, these efforts remain insufficient to fully eliminate childcare deserts [4]. Insufficient accessibility and high costs persist, particularly in regions most affected by economic hardship.

Economic and Policy Motivation

Addressing childcare deserts is not only a matter of social welfare but also of economic productivity. Expanding childcare infrastructure yields substantial economic returns: an investment of $1 billion in childcare facilities is estimated to create over 20,000 new jobs across New York State [7]. However, the shortage of facilities continues to constrain labor force participation and local economic activity. Therefore, the objective of this study is to propose optimal locations and expansion strategies that minimize cost while maximizing accessibility and fairness across communities.

3. Methods

3.1 Data Processing

Five datasets were used: `population.csv`, `employment.csv`, `avg_individual_income.csv`, `child_care_regulated.csv`, and `potential_locations.csv`. All datasets were cleaned and standardized prior to modeling.

ZIP codes were converted to 5-digit strings for consistency, and the population, employment, and income tables were merged by ZIP to form a unified dataset containing child populations by age, employment rate, income, and existing childcare capacity. Children aged 10–12 were approximated as 60% of the 10–14 population bucket:

$$P_z^{0-12} = P_z^{0-5} + P_z^{5-9} + 0.6\, P_z^{10-14}.$$

Employment rates were expressed as proportions (0–1), and missing or invalid values were handled conservatively.

Each ZIP code was then classified as either **high-demand** or **normal-demand**. High-demand ZIPs have employment $\geq 60\%$ or income $\leq$ \$60,000, and their required coverage ratio was set to $r_z = 0.5$; otherwise, $r_z = 1/3$.

Existing childcare capacity was aggregated from the regulated facility dataset. For each ZIP, we calculated total licensed capacity E_z^{tot} and 0–5 capacity $E_z^{0\text{–}5}$ (sum of infant, toddler, and preschool slots). At the facility level, expansion limits and baseline costs were derived as:

$$U_f = \min(1.2\, n_f, 500), \qquad B_f = 20{,}000 + 200\, n_f.$$

For spatial feasibility in later tasks, potential construction sites from `potential_locations.csv` were filtered to remove candidates within 0.06 miles of existing facilities, and mutually exclusive site pairs within 0.06 miles were recorded to impose location-spacing constraints.

Summary of preprocessing steps:

- Standardized and merged all datasets at the ZIP code level.
- Approximated 10–12 population as 60% of the 10–14 group.
- Classified ZIPs as high- or normal-demand based on employment and income.
- Aggregated existing childcare capacity and derived facility-level limits.
- Filtered candidate sites within 0.06 miles for spatial feasibility constraints.

3.2 General Assumptions

- **ZIP Code Classification:** Each ZIP code is categorized as either *high-demand* or *normal-demand*. High-demand areas meet at least one of the following: employment rate $\geq 60\%$ or

average income $\leq \$60,000$. These areas are considered childcare deserts if licensed capacity $\leq \frac{1}{2}$ of the local child population (ages 0–12). Normal-demand areas follow a $\frac{1}{3}$ threshold.

- **Facility Expansion Rules:** Existing facilities can expand by up to 20% of their current capacity n_f, or a maximum of 500 slots. Expansions exceeding 100% of n_f incur a baseline cost:

$$B_f = 20{,}000 + 200 \times n_f.$$

  Each additional slot costs \$220 on average (reflecting the \$200 per-slot cost plus a 10% overhead).

  Example: For a facility with 300 existing slots, the maximum 20% expansion allows 60 new slots. The total cost is:

$$220 \times 60 + (20{,}000 + 200 \times 300) = 13{,}200 + 80{,}000 = \$93{,}200.$$

- **New Construction:** New centers can be built in three discrete sizes (small, medium, large) as defined in Table 1. Because construction typically exceeds one year, expansions of newly built centers are excluded from this plan.

- **Coverage Requirement for Ages 0–5:** Each region must provide enough 0–5 slots to serve at least two-thirds (66.7%) of its local 0–5 population. Each 0–5 slot (new or expanded) requires an additional \$100 for age-specific equipment.

- **Budget and Objective:** The optimization minimizes total construction, expansion, and equipment costs under a total budget of \$100 million while ensuring full elimination of childcare deserts.

3.3 Modeling Framework Overview

To maintain consistency across the three tasks, this section introduces the sets, parameters, and variables used throughout the entire formulation.

Sets

- Z: set of ZIP codes.
- F: set of existing facilities.
- $S = \{\text{small}, \text{medium}, \text{large}\}$: types of new centers.

Parameters

- $P_z^{0\text{–}5}$: number of children aged 0–5 in ZIP z.
- $P_z^{\leq 12}$: number of children aged 0–12 in ZIP z.

- E_z^{tot}: current total licensed capacity in ZIP z.
- $E_z^{0\text{–}5}$: existing 0–5 childcare capacity (infant, toddler, preschool).
- r_z: coverage ratio (0.5 for high-demand, 1/3 for normal-demand ZIPs).
- $T_s \in \{100, 200, 400\}$: total number of slots per new facility type s.
- $K_s^{0\text{–}5} \in \{50, 100, 200\}$: maximum 0–5 slots assignable to new facility type s.
- $C_s^{\text{build}} \in \{65{,}000, 95{,}000, 115{,}000\}$: construction cost per new facility of type s.
- n_f: current total capacity of existing facility f.
- $U_f = \min(1.2n_f, 500)$: upper bound on possible expansion for facility f.
- $c^{\text{exp}} = 220$: average expansion cost per slot.
- $B_f = 20{,}000 + 200n_f$: baseline fixed cost if expansion reaches or exceeds 100% of n_f.

Decision Variables

- $y_{z,s} \in \mathbb{Z}_{\geq 0}$: number of new facilities of type s constructed in ZIP z.
- $x_f \in [0, U_f]$: number of expansion slots added to existing facility f.
- $w_f \in \{0, 1\}$: binary indicator equal to 1 if facility f expands by at least 100% of its current capacity.
- $n_z^{0\text{–}5} \geq 0$: number of new 0–5 childcare slots (either from new facilities or expansions) in ZIP z.

These definitions apply to all three tasks; later sections add fairness, location, or piecewise expansion features.

3.4 Mathematical Model — Task 1

Task 1 formulates the **idealized minimum-cost model** to eliminate all childcare deserts across New York State, assuming unrestricted expansion and construction capacity. The goal is to determine the least total funding required to satisfy the state's coverage requirements.

Decision Variables

- $y_{z,s} \in \mathbb{Z}_{\geq 0}$ — number of new childcare centers of size s to build in ZIP code z.
- $x_f \in [0, U_f]$ — number of additional slots added to existing facility f (expansion level).
- $w_f \in \{0, 1\}$ — binary variable equal to 1 if the expansion of facility f reaches or exceeds 100% of its current capacity, triggering a fixed baseline cost.

- $n_z^{0\text{–}5} \geq 0$ — number of new 0–5 childcare slots created in ZIP code z (either from expansions or new facilities).

Objective Function

The objective minimizes the total state funding required to eliminate all childcare deserts:

$$\min \quad \underbrace{\sum_{z\in Z}\sum_{s\in S} C_s^{\text{build}} y_{z,s}}_{\text{(a) New facility construction cost}} \quad + \quad \underbrace{\sum_{f\in F}\left(c^{\text{exp}} x_f + B_f w_f\right)}_{\text{(b) Expansion \& baseline trigger cost}} \quad + \quad \underbrace{100\sum_{z\in Z} n_z^{0\text{–}5}}_{\text{(c) Equipment cost for new 0–5 slots}}$$

where:

$$B_f = 20{,}000 + 200 n_f, \quad c^{\text{exp}} = 200.$$

Constraints

$$\begin{aligned}
&\text{(1) Desert elimination:} && E_z^{\text{tot}} + \sum_{f:\text{zip}(f)=z} x_f + \sum_{s\in S} T_s y_{z,s} \geq r_z P_z^{\leq 12}, && \forall z,\\
&\text{(2) Age 0–5 coverage:} && E_z^{0\text{–}5} + n_z^{0\text{–}5} \geq \tfrac{2}{3} P_z^{0\text{–}5}, && \forall z,\\
&\text{(3) Feasible 0–5 allocation:} && n_z^{0\text{–}5} \leq \sum_{s\in S} K_s^{0\text{–}5} y_{z,s} + \sum_{f:\text{zip}(f)=z} x_f, && \forall z,\\
&\text{(4) Expansion limits:} && 0 \leq x_f \leq U_f, && \forall f,\\
&\text{(5) Baseline trigger activation:} && x_f \geq n_f w_f, \quad x_f \leq (n_f - 1)(1 - w_f) + U_f w_f, && \forall f.
\end{aligned}$$

This model determines the minimal funding allocation necessary to satisfy all demand thresholds under unconstrained feasibility, serving as the baseline for subsequent tasks.

3.5 Mathematical Model — Task 2

Task 2 refines the baseline MILP by introducing **piecewise-linear expansion costs** and **location spacing constraints**. This version captures more realistic marginal costs for facility expansion and prevents clustering of new centers within close geographic proximity.

Additional Decision Variables

- $x_{f,1}, x_{f,2}, x_{f,3} \geq 0$ — number of expansion slots for facility f in each cost segment (0–10%, 10–15%, 15–20% of n_f).
- $y_{\ell,s} \in \{0, 1\}$ — binary variable indicating whether a new facility of type s is constructed at candidate location ℓ.

Objective Function

The objective remains consistent with Task 1, minimizing total expansion and construction costs, but the expansion cost is now divided into three piecewise segments with increasing marginal rates:

$$\min \quad \underbrace{\sum_{z\in Z}\sum_{s\in S} C_s^{\text{build}} y_{z,s}}_{\text{(a) New facility construction cost}} \quad + \quad \underbrace{\sum_{f\in F}\sum_{i=1}^{3} r_{f,i} x_{f,i}}_{\text{(b) Piecewise expansion cost}} \quad + \quad \underbrace{100\sum_{z\in Z} n_z^{0\text{–}5}}_{\text{(c) Equipment cost for 0–5 slots}}$$

where:

$$r_{f,1} = \frac{20{,}000 + 200n_f}{n_f}, \quad r_{f,2} = \frac{20{,}000 + 400n_f}{n_f}, \quad r_{f,3} = \frac{20{,}000 + 1000n_f}{n_f}.$$

Constraints

(1) Expansion decomposition: $x_f = x_{f,1} + x_{f,2} + x_{f,3}, \qquad \forall f,$

(2) Expansion limits by segment: $x_{f,1} \le 0.10n_f, \quad x_{f,2} \le 0.05n_f, \quad x_{f,3} \le 0.05n_f, \qquad \forall f,$

(3) Total expansion limit: $x_f \le 0.20n_f, \qquad \forall f,$

(4) Desert elimination: $E_z^{\text{tot}} + \sum_{f:\text{zip}(f)=z} (x_{f,1} + x_{f,2} + x_{f,3}) + \sum_{s\in S} T_s y_{z,s} \ge r_z P_z^{\le 12}, \qquad \forall z,$

(5) Age 0–5 coverage: $E_z^{0\text{–}5} + n_z^{0\text{–}5} \ge \frac{2}{3} P_z^{0\text{–}5}, \qquad \forall z,$

(6) Feasible 0–5 allocation: $n_z^{0\text{–}5} \le \sum_{s\in S} K_s^{0\text{–}5} y_{z,s} + \sum_{f:\text{zip}(f)=z} (x_{f,1} + x_{f,2} + x_{f,3}), \qquad \forall z,$

(7) Location spacing: $\sum_{s\in S} y_{\ell,s} + \sum_{s\in S} y_{k,s} \le 1, \qquad \forall(\ell,k)$ within 0.06 miles.

This refined formulation allows for heterogeneous marginal expansion costs and ensures that no two new childcare facilities are located within 0.06 miles of each other, enhancing the spatial realism and implementability of the optimization model.

3.6 Mathematical Model — Task 3

For the fairness-oriented extension of the project, the objective is to allocate facility expansion and construction decisions under a total budget of $100 million while ensuring equitable coverage across ZIP codes. This final stage builds upon the previous variables and constraints, introducing new elements for fairness, distance feasibility, and tiered expansion costs.

The model aims to:

1. Eliminate childcare deserts across all ZIPs,
2. Satisfy state policy thresholds for both age groups (0–5 and 6–12),

3. Ensure equitable access by minimizing regional disparities in coverage,
4. Respect physical spacing and realistic expansion feasibility limits.

Additional Sets and Parameters (new in Task 3)

- L: set of candidate construction sites; $L_z \subseteq L$ denotes sites within ZIP z.
- $T = \{\mathrm{S}, \mathrm{M}, \mathrm{L}\}$: facility types (small, medium, large).
- S_t: total capacity of facility type t.
- U_t: maximum number of 0–5 slots allowed for type t.
- C_t: construction cost of facility type t.
- $H_z \in \{0, 1\}$: high-need indicator ($H_z = 1$ if employment $\geq 60\%$ or income $\leq$ \$60,000).
- Child-care desert threshold:

$$R_z^{\text{desert}} = \left(\tfrac{1}{2}H_z + \tfrac{1}{3}(1 - H_z)\right) P_z^{0\text{–}12}.$$

- 0–5 coverage requirement:

$$R_z^{0\text{–}5} = \tfrac{2}{3} P_z^{0\text{–}5}.$$

New Decision Variables

- $x_f^{0\text{–}5}$: portion of expansion x_f dedicated to 0–5 age slots at facility f.
- $y_{l,t} \in \{0, 1\}$: 1 if a new facility of type t is built at location l.
- $s_{l,t}$: total capacity of a newly built facility of type t at l.
- $u_{l,t}$: 0–5 capacity among the total $s_{l,t}$ slots.
- $\mathrm{cov}_z^{0\text{–}5}$, $\mathrm{cov}_z^{5\text{–}12}$: achieved coverage ratios for each age group in ZIP z.
- m, M: lower and upper fairness bounds across ZIPs.

Objective Function

Maximize the population-weighted social coverage index:

$$\max \sum_{z \in Z} \left(\frac{2}{3} \,\mathrm{slots}_z^{0\text{–}5} + \frac{1}{3} \,\mathrm{slots}_z^{5\text{–}12} \right).$$

Capacity Balance Constraints

$$\begin{aligned}
\text{slots}_z &= \sum_{f \in F_z} (n_f + x_f) + \sum_{l \in L_z} \sum_{t \in T} s_{l,t}, \\
\text{slots}_z^{0\text{–}5} &= \sum_{f \in F_z} (n_f^{0\text{–}5} + x_f^{0\text{–}5}) + \sum_{l \in L_z} \sum_{t \in T} u_{l,t}, \\
\text{slots}_z^{5\text{–}12} &= \text{slots}_z - \text{slots}_z^{0\text{–}5}.
\end{aligned}$$

Policy and Desert-Elimination Constraints

$$\begin{aligned}
\text{slots}_z &\geq R_z^{\text{desert}}, && \forall z, \\
\text{slots}_z^{0\text{–}5} &\geq R_z^{0\text{–}5}, && \forall z.
\end{aligned}$$

Fairness Constraints

Define the overall coverage rate:

$$\text{cov}_z = \tfrac{2}{3}\text{cov}_z^{0\text{–}5} + \tfrac{1}{3}\text{cov}_z^{5\text{–}12}.$$

All ZIP codes must remain within a narrow fairness band:

$$m \leq \text{cov}_z \leq M, \qquad \forall z, \qquad M - m \leq 0.1.$$

Budget and Cost Structure

Expansions follow a three-tier piecewise-linear cost structure (0–10%, 10–15%, 15–20% of existing n_f):

$$x_f = x_{f,1} + x_{f,2} + x_{f,3}, \qquad x_{f,1} \leq 0.10 n_f,\ x_{f,2} \leq 0.05 n_f,\ x_{f,3} \leq 0.05 n_f.$$

$$r_{f,1} = \frac{20000 + 200 n_f}{n_f}, \quad r_{f,2} = \frac{20000 + 400 n_f}{n_f}, \quad r_{f,3} = \frac{20000 + 1000 n_f}{n_f}.$$

Total budget constraint:

$$\sum_f (r_{f,1} x_{f,1} + r_{f,2} x_{f,2} + r_{f,3} x_{f,3}) + \sum_{l,t} C_t y_{l,t} + 100 \left(\sum_f x_f^{0\text{–}5} + \sum_{l,t} u_{l,t} \right) \leq 100{,}000{,}000.$$

Distance Constraints

For any candidate pair (p, q) within 0.06 miles:

$$\sum_t y_{p,t} + \sum_t y_{q,t} \leq 1.$$

4. Results

4.1 Results — Task 1

Regarding the idealistic model (Task 1), the optimization determined for every ZIP code the facilities required to eliminate childcare deserts. These facilities could either be newly built or expanded. In the case of new construction, the model also determined the optimal size (small, medium, or large), while for expansions it specified the number of slots to be added.

The first model was executed over approximately 1,825 ZIP codes. Among these, 1,720 were initially identified as childcare deserts, and all were eliminated after optimization. This means that the model's objective, complete childcare desert elimination, was achieved in the idealized scenario.

The model determined that **70 new facilities** should be built across New York State: **35 small**, **25 medium**, and **10 large** centers. The new capacity created through construction was approximately **170,000 slots**, while the total capacity added through expansion reached **40,000 slots**. Within these, about **55,000 slots** were assigned to the 0–5 age group. In total, **210,000 new slots** were created statewide, roughly 80% from new construction and 20% from expansion.

The new facilities were primarily located in areas previously classified as major childcare deserts, while the expansions were concentrated in dense, urban ZIP codes where large facilities already existed. The total cost (objective value) of this model amounted to approximately **$311 million**.

Since there were no distance or candidate-location constraints in this first scenario, the model achieved full elimination efficiently. It is also observed that the model favored **medium-sized facilities**, as they presented the lowest cost per slot (approximately **$1,800 per slot**), providing an optimal cost-capacity balance.

4.2 Results — Task 2

In Task 2, the same ZIP codes and initial desert classifications were analyzed, but with added spatial and piecewise-expansion constraints. As the model incorporated more realistic assumptions, a small number of childcare deserts remained after optimization.

After running the model, approximately **30–40 ZIP codes** remained classified as deserts. The model selected **65 new sites** for childcare facilities: **25 small**, **25 medium**, and **15 large** centers. The capacity added by new construction was approximately **150,000–160,000 slots**, and the capacity from expansions reached around **30,000–40,000 slots**, for a total added capacity of about **190,000–200,000 slots**.

The total cost remained similar to that of Task 1, but the composition shifted slightly. The model prioritized expanding existing facilities by about **25–30%**, relative to the 20% expansion limit observed in Task 1. However, about **70–75%** of the additional capacity still came from new construction.

Most new facilities were located in ZIP codes with no existing childcare infrastructure or very high demand ratios. As expected, the model produced more small and medium-sized facilities, consistent with the cost structure assumptions.

Overall, **over 97% of all childcare deserts were eliminated**, with the remaining 3% corresponding to ZIP codes where no suitable candidate sites were available. The newly constructed centers were concentrated mainly in **Western and Central New York**, reflecting the regions with the most severe accessibility gaps.

4.3 Results — Task 3

In Task 3, the model became significantly more restrictive. A total budget cap of **$100 million** was imposed, and the objective function was changed from minimizing cost to maximizing overall coverage while enforcing fairness constraints across ZIP codes.

Under these additional conditions, the model was found to be **infeasible**. The fairness constraint forced coverage levels to be nearly equal across all ZIP codes while simultaneously restricting spending to $100 million. This created conflicting requirements, reducing disparities across regions increases costs nonlinearly, as it forces the construction of large facilities even in small or low-demand ZIP codes.

Why the Model Is Infeasible

The fairness-augmented model was found to be infeasible under the provided dataset and budget. Three necessary conditions fail:

(a) **Budget Lower Bound Violation:** Even under the cheapest combination of expansion and construction, the minimum cost to satisfy all deserts exceeds the $100 million cap.

(b) **Physical Capacity Upper Bound:** Several ZIP codes lack sufficient candidate sites or allowable expansion capacity to reach their desert thresholds, violating

$$\mathrm{Cap}_z^{\max} < R_z^{\text{desert}} \quad \text{or} \quad \mathrm{Cap}_z^{\max} < R_z^{0\text{–}5}.$$

(c) **Fairness Band Violation:** High-population ZIPs have low achievable coverage, while smaller ZIPs easily surpass thresholds, making the required $M - m \leq 0.1$ band impossible to maintain.

These findings confirm that the fairness-constrained MILP cannot achieve feasibility under the current assumptions. For future extensions, it is recommended to either **increase the budget** or **relax the fairness bound (e.g., allow $M - m \leq 0.2$)** to achieve feasible and equitable solutions that reflect real-world trade-offs.

5. Conclusion

In the idealized expansion and construction model (**Task 1**), all childcare deserts across New York State were successfully eliminated at a minimum cost of approximately **$311 million**, resulting in the addition of **210,000 total childcare slots**.

In the more realistic location-based model (**Task 2**), the vast majority of deserts (over 97%) were eliminated, with a similar total cost but slightly reduced capacity (around **170,000–190,000 slots**), due to the impact of spatial and piecewise-expansion constraints.

However, in the fairness-oriented model (**Task 3**), the combination of strict equity requirements and limited budget resulted in infeasibility. This demonstrates that while equity in access is a desirable policy goal, it comes at a significant financial cost and may require relaxed fairness bands or increased funding to be realistically achieved.

Overall, the results indicate that the **elimination of childcare deserts in New York State is achievable at a reasonable cost** and would substantially improve accessibility for families. Expanding existing facilities remains a cost-effective strategy, while new construction is essential to serve underserved regions. Together, these insights provide a robust, data-driven framework for policymakers to plan equitable and efficient childcare infrastructure investments across the state.

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